Chapter 1

How Cameras Work

Let's say it is a sunny day and you are standing in a field a few meters from a cow. You use the camera on your phone to take a picture of the cow. How does that whole process work?

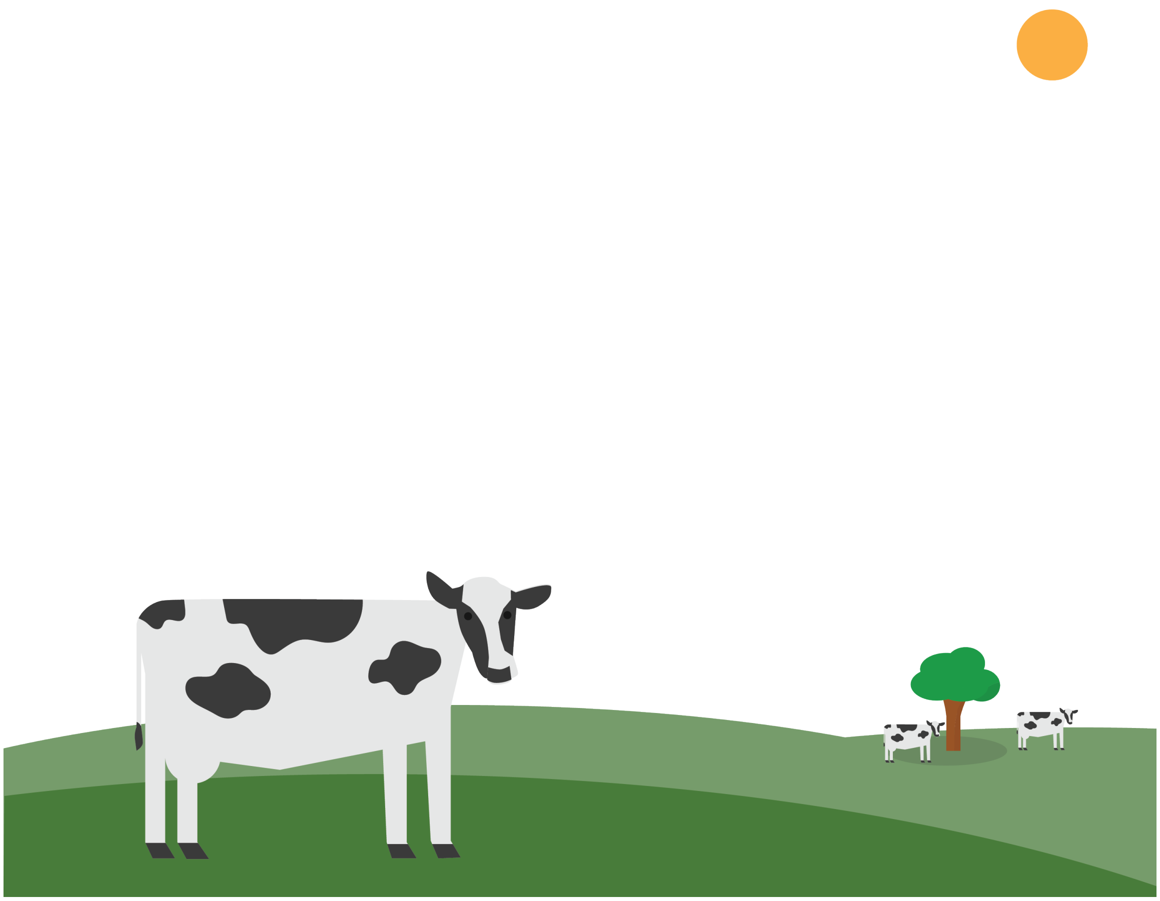

Figure 1.1: A cow in a field.

1.1 The Light That Shines On the Cow

The sun is a sphere of hot gas. About 70% of the gas is hydrogen. About 28% is helium. There's also a little carbon, nitrogen, and oxygen.

Gradually, the sun is converting hydrogen into helium through a process known as "nuclear fusion" (which we will be discussing more in a future chapter). A large amount of heat is created in this process. This heat makes the gases glow.

How does heat make things glow? The heat pushes the electrons into higher orbitals. When they come back down to a lower orbital, they release a photon of energy, which travels away from the atom as an electromagnetic wave. Recall that when we talked about reflections, we established that light is a constant stream of particles we call photons. These photons travel at the speed of light.

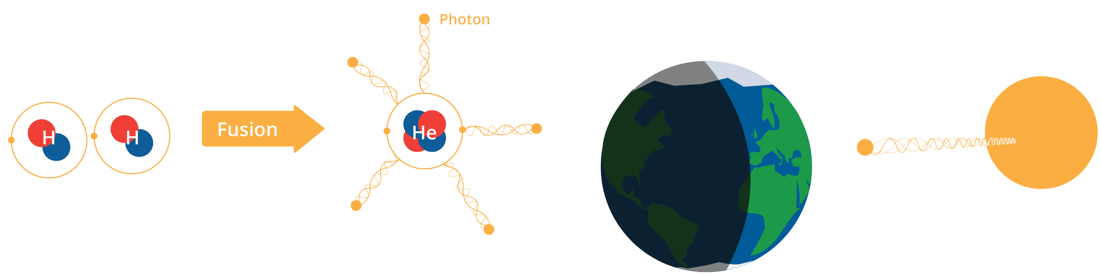

Figure 1.2: Photons are released when the sunlight hits the cow.

Heat is not the only way to push the electrons into a higher orbital. For example, a fluorescent lightbulb is filled with gas. When we pass electricity through the gas, its electrons are moved to a higher orbital. When they fall back to a lower orbital, light is created.

What is the frequency of the wave that the photon travels on? Depending on what orbital it falls from and how far it falls, the photon created has different amounts of energy. The amount of energy determines the frequency of the electromagnetic wave, and ultimately, its color!

In the sun, there are several kinds of molecules and each has a few different orbitals that the electrons can live in. Thus, the light coming from the sun is made up of electromagnetic waves of many different frequencies.

We can see some of these frequencies as different colors, but some are invisible to humans, such as ultraviolet and infrared.

1.2 Light Hits the Cow

When these photons from the sun hit the cow, the hide and hairs of the cow will absorb some of the photons. These photons will become heat and make the cow feel warm. Some of the photons will not be absorbed – they will leave the cow. When you say "I see the cow," what you are really saying is "I see some photons that were not absorbed by the cow."

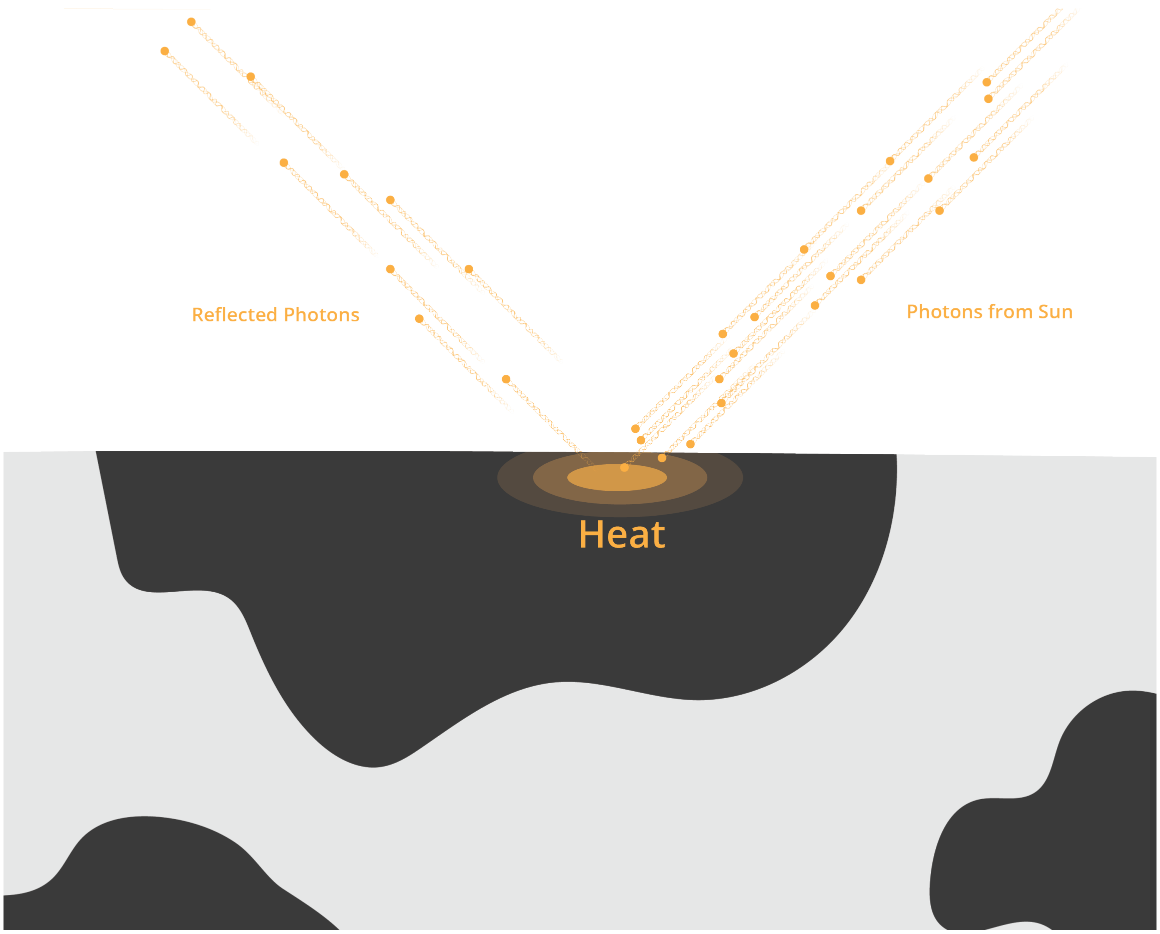

Figure 1.3: Photons hitting the cow create a heated spot, where some are absorbed.

Different materials absorb different amounts of each wavelength. A plant, for example, absorbs a large percentage of all blue and red photons that hit it, but it absorbs only a small percentage of the green photons that hit it. Thus, we say "That plant is green."

White things absorb very small percentages of photons of any visible wavelength. Black things absorb very *large* percentages of photons of any visible wavelength. That is why, on a hot summer day, a black car with black seats and interior will heat up on the inside much hotter than a white car.

Before we go on, let's review: The sun creates photons that travel as electromagnetic waves of assorted wavelengths to the cow. Many of those photons are absorbed, but some are not. Some of those photons that are not absorbed go into the lens of our camera.

1.3 Pinhole camera

The simplest cameras have no lenses. They are just a box. The box has a tiny hole that allows photons to enter. The side of the box opposite the hole is flat and covered with film or some other photo-sensitive material.

The photons entering the box continue in the same direction they were going when they passed through the hole. Thus, the photons that entered from high hit the back wall at a low point. The photons that came from the left hit the back wall on the right. This is how the image is projected onto the back wall, rotated 180 degrees; what was up is down, what was on the left is on the right.

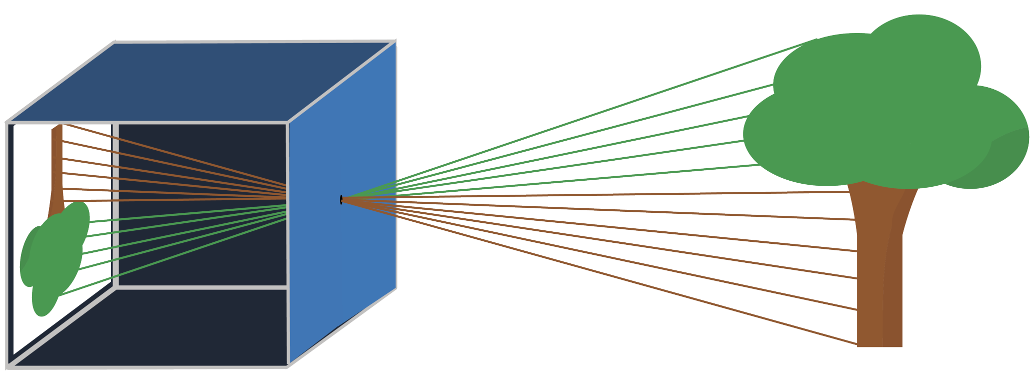

Figure 1.4: A pinhole camera flips the image it "sees" by 180°.

Exercise 1 Height of the image

Let's say that the pinhole is exactly the same height as the shoulder of the cow, and that the shoulder is directly above one hoof. This means the pinhole, the shoulder, and the hoof form a right triangle.

Now, let's say that the camera is being held perpendicular to the ground. The pinhole, the image of the shoulder, and the image of the hoof on the back wall of the camera now also form a right triangle.

These two triangles are similar.

The shoulder is 2 meters from the hoof. The cow is standing 3 meters from the camera. The distance from the pinhole to the back wall of the camera is 3 cm. How tall is the image of the cow on the back wall of the camera?

Working Space

Answer on Page 11

1.4 Lenses

Now, a quick review: A photon leaves the sun in some random direction. It travels 150 million km from the sun and hits a cow. It is not absorbed by the cow, and heads off in a new direction. It passes through the pinhole and hits the back wall of the camera. That seems incredibly improbable, right?

It actually is relatively improbable, especially if there isn't a lot of light — like you are taking the picture at dusk. To increase the odds, we added a *lens* to the camera. If you focus a lens on a wall and you draw a dot on that wall, the lens is designed such that all the photons from the dot that hit the lens get redirected to the same spot on the back wall of the camera — regardless of which path it took to get to the lens.

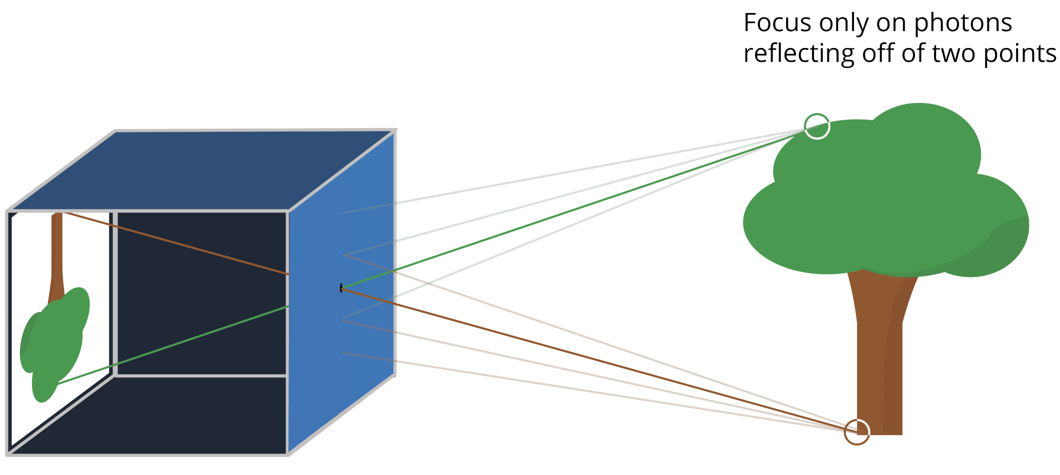

Figure 1.5: A pinhole with a lens allows you to choose the points it reflects.

Note that the image still gets flipped. There is a *focal point* that all the photons pass through.

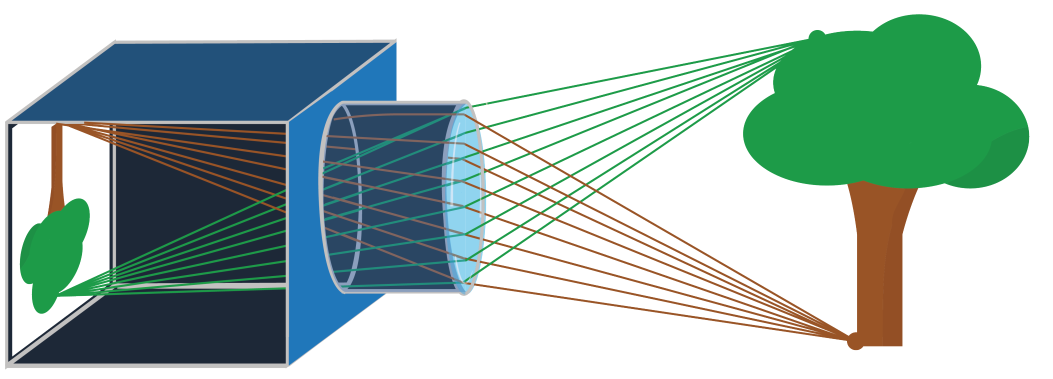

Figure 1.6: A lens still flips the image, but introduces a focal point.

The distance from the lens to its focal point is called the lens's *focal length*. Telephoto lenses, that let you take big pictures of things that are far away, have long focal lengths.

Wide-angle lenses have short focal lengths.

1.5 Sensors

The camera on your phone has a sensor on the back wall of the camera. The sensor is broken up into tiny rectangular regions called pixels. When you say a sensor is 6000 by 4000 pixels (most common ratio for photography), we are saying the sensor is a grid of 24,000,000 pixels: 6000 pixels wide and 4000 pixels tall.

In the sensor, each individual pixel is hit by an incoming photon. If the photon's energy (recall $E = hf$) is less than the Work Function Φ to dislodge an electron, the light simply passes through or reflects, and the pixel remains *dark*, or what we call *black*. If $hf \geq \Phi$, the photon is absorbed, and its energy is transferred to an electron, which *jumps* into the conduction band, appearing like *white* to us. Camera sensors then convert their pixels into a binary mapping, using voltage measurements.

Each pixel has three types of cavities that take in photos. One of the cavities measures the amount of short wavelength light, like blues and violets. One of the cavities measures the long wavelength light, like reds and oranges. One of the cavities measures the intensity of wavelengths in the middle, like greens. Thus, if your camera has a *resolution* of 6000×4000, the image is 24,000,000 pixels yields three numbers: intensity of long wavelength, mid wavelength, and short wavelength light, for a total of 72,000,000 numbers. We call these numbers "RGB" for Red, Green, and Blue. The RGB values range from $0 - 255$ for each channel. We will talk more about this when hexadecimal is introduced.

Exercise 2 Camera Pixels

A digital camera sensor is made of a material with a work function of $\Phi = 1.1$ electronvolts. If the sensor is exposed to monochromatic light with a wavelength of $\lambda = 1200$ nm, what does the resulting image look like? Why?

Hint: $E = \frac{1240}{\lambda}$ where λ is in nanometers and E is in electronvolts (eV).

Working Space

Answer on Page 11

1.6 Aspect Ratio

When you are buying a monitor, TV, or even smartphone, you may hear the term **aspect ratio**. The aspect ratio is a set of two numbers representing the ratio between the width and the height. Conventionally, the width comes first, separated by a colon, and then the height:

$$\text{width} : \text{height}$$

Some of the most common aspect ratios are:

- $16:9$ (known as Widescreen)
- $17:9$
- $9:16$
- $4:3$ (known as Fullscreen)
- $4:5$
- $3:2$
- $1:1$ (perfect square, like on a smartwatch)

As we talked about above, the aspect ratio doesn't describe the actual size of the screen or image. Instead, it describes the shape. A movie theatre screen can have the same aspect ratio as your monitor, even if the screen is 360 inches and the monitor is 24 inches.

Aspect ratio describes the proportional relationship between width and height. Resolution describes the total number of pixels used to represent an image.

For example, both of the following resolutions have a $16:9$ aspect ratio:

$$1280 \times 720$$
$$1920 \times 1080$$

Mathematically, the aspect ratio is a dimensionless quantity. Since both width and height are measured in the same units (such as centimeters, inches, or pixels), the units cancel when forming the ratio. This means aspect ratio describes proportion rather than physical measurement. Most often, we determine the aspect ratio of a digital display by comparing the number of horizontal pixels to vertical pixels.

For example:

$$1920 \times 1080 \quad \Rightarrow \quad \frac{1920}{1080} = \frac{16}{9} = 16 : 9$$

Digital cameras and smartphone cameras capture images using a rectangular sensor. The shape of this sensor determines the default aspect ratio of the image. For example:

- Many digital cameras use a $3 : 2$ sensor.
- Micro Four Thirds cameras use $4 : 3$.
- Most video recording uses $16 : 9$.

Important: when you change the aspect ratio setting on a camera, the device does not change the sensor itself. Instead, it crops part of the image to match the selected ratio. Cropping reduces the number of recorded pixels, which may reduce file size. However, actual file size also depends on compression methods used by the device.

Human vision has a wider horizontal field of view than vertical field of view. Because of this, wider aspect ratios such as $16 : 9$ often feel more natural and immersive. This is one reason modern TVs and cinema formats favor wider screen shapes. However, with the rise of cellphones, we have switched to consistently using portrait or vertical mode, and video aspect ratios are more consistently becoming $9 : 16$. This allows for consistency in device quality, and commonly used for vertical videos, like in short-form content, compared to traditional widescreen videos, such as Youtube or traditional horizontal video capturing.

1.6.1 Printing

You can imagine this may cause a problem for printing out photos. Most printing paper is 8.5 inches $\times$ 11 inches or 8 inches $\times$ 10 inches. This causes an issue, as that fits a $8.5 : 11 \iff 17 : 22$ or $4 : 5$ aspect ratio, respectively. We combat this by using very standardized printing sizes (in inches):

- $6 \times 4 \to 3 : 2$,
- $7 \times 5 \to 7 : 5$,
- $10 \times 8 \to 5 : 4$,
- and, rarely, $8.5 \times 11 \to 1.29 : 1$

Going to any framing or printing store will give you a very standardized set of sizes, or some multiple of these.

Most smartphones use $4 : 3$ or $3 : 2$, which will work for 4×6 inches or 8×6 inches, respectively.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 5)

The two triangles are similar; one is 2 m and 3m, the other is x cm and 3 cm.

The image of the cow is 2 cm tall.

Answer to Exercise 2 (on page 7)

The pixels will remain dark, or black, because the energy of the incoming photons is less than the work function of the sensor material. Calculating $E = \frac{1240}{1200} = 1.033\,\text{eV}$, which is less than the work function of 1.1 eV.

Index